

LABORATORY OBSERVATION

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Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 5-4

1. TITLE

DOM Demonstration

2. AIM

To explain and demonstrate the Document Object Model (DOM) and show how JavaScript can dynamically change the content, style, and attributes of HTML elements.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the structure and purpose of the DOM.
 - To access HTML elements using JavaScript.
 - To dynamically change the content of an HTML element.
 - To dynamically change the style of an HTML element.
 - To modify an HTML attribute using JavaScript.
 - To use an event listener to perform DOM changes when a button is clicked.
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4. THEORY

The Document Object Model (DOM) is a programming interface that represents an HTML document as a tree of objects. Each HTML element is represented as a DOM element that JavaScript can access and modify.

JavaScript can select elements using methods such as `getElementById()`. After an element is selected, its content can be changed using `textContent`, its appearance can be changed through the `style` property, and its HTML attributes can be changed using methods such as `setAttribute()`.

An event listener allows JavaScript to execute a function when a specified event occurs. In this demonstration, a click event on the button triggers all the DOM changes without refreshing the webpage.

5. ALGORITHM / PROCEDURE

1. Create an HTML page containing a heading, status text, message, image, attribute status, and button.
2. Assign unique IDs to the elements that will be modified.
3. Use JavaScript `getElementById()` to select the required DOM elements.

4. Attach a click event listener to the button.
 5. Change the message content using `textContent`.
 6. Change the heading color and font size using the `style` property.
 7. Change the message background color using the `style` property.
 8. Change the image alt attribute using `setAttribute()`.
 9. Update the status, attribute message, and button text.
 10. Run the page in a browser and verify the original and changed DOM states.
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6. PROGRAM

index.html

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>DOM Demonstration</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

  <div class="container">

    <h1 id="heading">DOM Demonstration</h1>

    <p id="status">Status: Original DOM</p>

    <div id="message" class="message">

      Content on Content Creation Content creation is the process of

      developing and sharing valuable, relevant, and engaging material

      to connect with a specific audience.
```

style.css

```
body {  
  
    font-family: Arial, sans-serif;  
  
    background-color: #ffffff;  
  
    margin: 0;  
  
    padding: 50px;  
  
}  
  
.container {  
  
    width: 620px;  
  
    padding: 25px;  
  
    border: 1px solid #aaa;  
  
}  
  
h1 {  
  
    font-size: 40px;  
  
    margin-bottom: 25px;  
  
}
```

6. PROGRAM (CONTINUED)

script.js

```
const heading = document.getElementById("heading");

const status = document.getElementById("status");

const message = document.getElementById("message");

const image = document.getElementById("demoImage");

const attributeStatus =

    document.getElementById("attributeStatus");

const button = document.getElementById("changeButton");

button.addEventListener("click", function () {

    // Step 1: Change the content

    message.textContent =

        "Content Creation: Content creation is the process of developing " +

        "and sharing valuable, relevant, and engaging material to connect " +

        "with a specific audience. It can take many forms, such as articles, " +

        "videos, podcasts, infographics, or social media posts. It serves " +

        "purposes like educating, entertaining, or inspiring people. " +

        "Effective content creation involves understanding the audience's " +

        "needs, planning strategically, and delivering information in a " +

        "clear and creative way.";
```

7. CODE EXPLANATION

Code	Explanation
getElementById()	Selects HTML elements from the DOM using their IDs.
addEventListener("click", ...)	Runs the function when the button is clicked.
textContent	Changes the text content of the selected HTML element.
style.color	Changes the text color of the heading.
style.fontSize	Changes the heading font size.
style.backgroundColor	Changes the background color of the message.
setAttribute("alt", ...)	Changes the alt attribute of the image.
button.textContent	Changes the text displayed on the button.

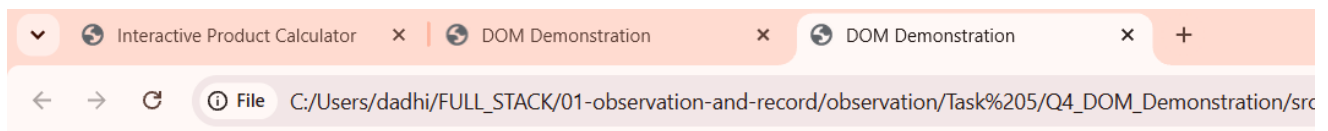
8. TEST CASES AND EXECUTION DATA

Test Case	Action / Input	Expected Result	Actual Result	Status
TC-01	Load webpage	Original DOM state is displayed.	Status: Original DOM displayed.	Pass

Test Case	Action / Input	Expected Result	Actual Result	Status
TC-02	Click Start DOM Changes	Content, style, and image attribute are changed.	Status: Content, Style and Attribute Changed.	Pass
TC-03	Verify image alt attribute	Alt attribute changes to Updated Image.	Updated Image displayed.	Pass
TC-04	Verify button text	Button text changes after DOM update.	DOM Changes Completed displayed.	Pass

9. OUTPUT

Output 1: Original DOM State



DOM Demonstration

Status: Original DOM

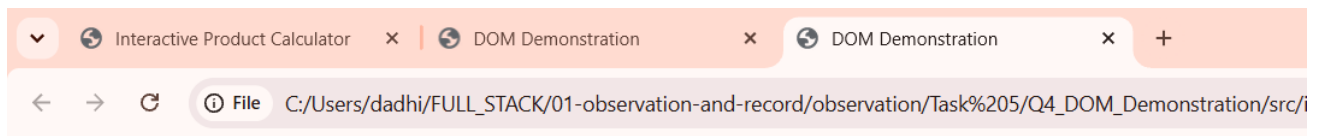
Content on Content Creation Content creation is the process of developing and sharing valuable, relevant, and engaging material to connect with a specific audience. It can take many forms—such as articles, videos, podcasts, infographics, or social media posts—and digital world, it plays a vital role in building brand awareness, fostering trust, and driving meaningful engagement. When done consistently and thoughtfully, content creation becomes a powerful tool for communication and growth.



Image alt attribute: Original Image

Start DOM Changes

Output 2: DOM After JavaScript Changes



DOM Demonstration

Status: Content, Style and Attribute Changed

Content Creation: Content creation is the process of developing and sharing valuable, relevant, and engaging material to connect with a specific audience. It can take many forms, such as articles, videos, podcasts, infographics, or social media posts. It serves purposes like educating, entertaining, or inspiring people. Effective content creation involves understanding the audience's needs, planning strategically, and delivering information in a clear and creative way.



Image alt attribute: Updated Image

DOM Changes Completed

10. RESULT

Thus, the DOM was explained and demonstrated using HTML, CSS, and JavaScript. The supplied screenshots verify the original DOM state and the changes made to content, style, image attribute, status text, and button text after the button click.

How this would be submitted

Source Code

```
Q4_DOM_Demonstration/  
├── src/  
│   ├── index.html  
│   ├── style.css  
│   ├── script.js  
│   └── content.jpg  
└── README.md
```